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Cite this as: *BMJ* 2026;392:s133
http://doi.org/10.1136/bmj.s133
Published: 22 January 2026

We must no longer accept waiting years to understand disease burden

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One of the most important advances to emerge from the covid-19 pandemic is the ability to measure population health at unprecedented scale and speed. Our study of time trends for long term conditions, published in *The BMJ* (doi:10.1136/bmj-2025-086393),[1] shows how electronic health record data routinely captured across primary and secondary care can reveal dynamic changes in disease patterns as they occur. The wider opportunities extend far beyond the pandemic, with implications for public health surveillance, research, and healthcare delivery.

To plan services and allocate resources healthcare systems have traditionally relied on static estimates of disease burden that are often published years after data collection. Measures of disease incidence and prevalence are treated as fixed truths, rather than estimates that vary substantially over time and by demographic and geography. The data presented in our study show how rapidly diagnostic activity can change and how variable the responses to external pressures, such as public health emergencies, can be. Diagnosis rates did not fall and rebound uniformly across multiple diseases—some diseases recovered quickly, others lagged, and some may never return to prepandemic trajectories. Without access to timely data, these changes can remain hidden until it is too late to respond.

Importantly, the data infrastructure now exists to track changes in disease patterns as they emerge. Health data platforms such as OpenSAFELY make it possible to analyse routinely collected health data for tens of millions of people in near real time. Privacy safeguards mean that no individual patient data are visible to researchers, and all analysis code and platform activity are shared publicly.[2] These capabilities are transformative. By enhancing our understanding of how common diseases are and how they vary by population and over time, we can better model workforce needs, target prevention efforts, and predict future demand. This enables policymakers and clinicians to act earlier, directing services and resources to where they are needed most.

The same data infrastructure provides an opportunity to transform how we monitor and deliver improvements in the quality of care that patients receive. Each year, countless clinical guidelines are published by national and international bodies. Despite this we often have little insight into whether their evidence based recommendations are being adopted into routine care. Existing approaches, such as clinical audit, rely heavily on manual data collection, which is time consuming and biased towards units with greater resources.[3] Local audits provide snapshots of performance that are rarely published, while national audit programmes cover only a fraction of diseases[4] and often miss large numbers of patients altogether.[5]

We must move away from manual data collection and transform how we monitor care quality at scale. Clinically coded information on diagnoses, prescriptions, and investigations can be analysed through platforms such as OpenSAFELY to assess care quality across entire populations without adding burden to clinicians. Importantly, this needs to be done continuously rather than as a retrospective snapshot. Public facing dashboards that are already available can provide timely feedback on whether evidence based therapies are reaching patients equitably, whether care gaps are widening, and where targeted support is needed.[6]

These capabilities are particularly useful when it comes to designing and testing strategies to optimise patient care. Guidelines alone are rarely sufficient to change practice, and the process of change is often slow.[7] Capacity and workforce pressures, competing priorities, and entrenched workflows are common barriers to implementation. Targeted strategies are therefore essential to drive change. Some of the most effective approaches incorporate digital solutions, such as electronic health record prompts and templates, that reduce burden on clinicians. The effectiveness of health interventions can now be evaluated using data enabled trials that require no manual data collection whatsoever.[8,9] This reduces costs enormously, while improving the generalisability of the findings, and opens the door to rapid, iterative evaluations of quality improvement interventions.

The pandemic forced healthcare systems to adapt at unprecedented speed. We now have the opportunity to move decisively from analogue to digital ways of understanding and improving care. This requires continued investment in data infrastructure, reusable analytical code, and transparent frameworks. Data driven monitoring of disease burden and care quality must be treated as core public health functions, not as optional research activities. Waiting years for retrospective analyses to evaluate changes in disease burden or care quality is unacceptable when better alternatives exist.

Competing interests: MDR has received honorariums from AbbVie, Biogen, Galapagos, Johnson & Johnson, Lilly, Menarini, Novartis, Pfizer, UCB, Viforpharma and grant funding from Sandoz UK. JBG has received honorariums from Abbvie, Biovitrum, BMS, Celgene, Chugai, Galapagos, Gilead, Janssen, Lilly, Novartis, Pfizer, Roche, Sanofi, Sobi, UCB, and grant funding from Sandoz UK.

Provenance and peer review: Commissioned, not externally peer reviewed.

1 Russell MD, Schaffer A, Bechman K, etal. Time trends in newly recorded diagnoses of 19 long term conditions before, during, and after the covid-19 pandemic: population based cohort study in England using OpenSAFELY. *BMJ* 2026;392:e086393. doi: 10.1136/bmj-2025-086393.
2 OpenSAFELY. About OpenSAFELY. 2025. https://www.opensafely.org/about/.
3 Glew S, Sornalingam S, Crossman T. Audit: time to review the cycle. *Br J Gen Pract* 2014;64:-7. doi: 10.3399/bjgp14X682573 pmid: 25452509

4 Healthcare Quality Improvement Partnership. The national clinical audit programme. 2026. https://www.hqip.org.uk/a-z-of-nca/.

5 Yates M, Bechman K, Dennison EM, etal. Data quality predicts care quality: findings from a national clinical audit. *Arthritis Res Ther* 2020;22:. doi: 10.1186/s13075-020-02179-y pmid: 32303251

6 OpenSAFELY. OpenSAFELY reports. 2025. https://reports.opensafely.org/

7 Cabana MD, Rand CS, Powe NR, etal. Why don't physicians follow clinical practice guidelines? A framework for improvement. *JAMA* 1999;282:-65. doi: 10.1001/jama.282.15.1458 pmid: 10535437

8 Sydes MR, Barbachano Y, Bowman L, etalData Enabled Trials Group Workshop Group members. Realising the full potential of data-enabled trials in the UK: a call for action. *BMJ Open* 2021;11:e043906. doi: 10.1136/bmjopen-2020-043906 pmid: 34135032

9 Ainsworth B, Horwood J, Walter SR, etalOpenSAFELY Collaborative. Implementing Germ Defence digital behaviour change intervention via all primary care practices in England to reduce respiratory infections during the COVID-19 pandemic: an efficient cluster randomised controlled trial using the OpenSAFELY platform. *Implement Sci* 2023;18:. doi: 10.1186/s13012-023-01321-z pmid: 38049846